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1 // Pin assignments
2 const int button1 = 9;
3 const int button2 = 10;
4 const int button3 = 11;
5 // Variables to represent state
6 const int BEEPING = 0;
7 const int WIGGLING = 1;
8 const int SINGING = 2;
9 // State variable and initial value
10 int state = BEEPING;
11
12 // State: Beeping
13 void beep() {
14     // Beeping action
15     Serial.println("Beep beep");
16
17     // Transitions to other states
18     if (digitalRead(button1)) {
19         state = WIGGLING;
20     } else if (digitalRead(button2)) {
21         state = SINGING;
22     }
23 }
24
25 // State: Wiggling
26 void wiggle() {
27     // Wiggling action
28     Serial.println("Wiggle wiggle");
29
30     // Transitions to other states
31     if (digitalRead(button3)) {
32         state = BEEPING;
33     } else if (digitalRead(button2)) {
34         state = SINGING;
35     }
36 }
37
38 // State: Singing
39 void sing() {
40     // Singing action
41     Serial.println("Lalalalalala");
42
43     // Transitions to other states
44     if (digitalRead(button1)) {
45         state = WIGGLING;
46     } else if (digitalRead(button3)) {
47         state = BEEPING;
48     }
49 }

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51 // Finally ready to go: typical setup function
52 void setup() {
53     Serial.begin(9600);
54     pinMode(button1, INPUT);
55     pinMode(button2, INPUT);
56 }
57
58 // Keep checking state variable,
59 // and perform appropriate action
60 void loop() {
61     if (state == BEEPING) {
62         beep();
63     } else if (state == WIGGLING) {
64         wiggle();
65     } else if (state == SINGING) {
66         sing();
67     } else {
68         Serial.print("Bad state: ");
69         Serial.println(state);
70     }
71 }

```

void jump() {
 // jumping action
 Serial.println("—");
 // Transitions
 == ... state = JUMP
 }

1. What are the three different actions that this program can do?

beep beep la la la la
wistle wistle

2. In this program, I kept things simple by just using a `Serial.println` for each action to announce the action. For each of the three actions, explain what an Arduino might really do. In other words, "beep" probably wouldn't actually print out the words "Beep beep". Specify for each action what hardware we would likely be using, and what might really be done.

beep → beeper
wistle → pos servo
singing - multifreq beeper
(laptop play?)

3. What line number actually calls the function beep?

3. 62

4. When this program starts, before any buttons are pressed, what does it do? Explain, and show sample output below.

beep beep

5. Suppose that after the program is running for a little while, a user holds down button 1 (i.e., the button connected to pin button1) for a short while. What line of code detects this?

5. 18

6. Following on the previous question, after the button 1 has been detected, what global variable is changed as a result, and to what value?

state → WIGGLING

7. Following on the previous question, what effect does changing that global variable have? Specifically, indicate what lines of code inside loop end up behaving differently as a result, and how.

wiggle wiggle shh

instead of 61, goes to 63

8. Show what sample output might look like for this program if after it starts running, button 1 is pushed after a short while, then button 3, then button 2, then button 2 again, then button 1.

beep
wiggle
beep
tala
tala
wiggle

9. I wish to add a fourth state, which would result in the Arduino printing "jump jump". Write that function on the first page, on the bottom right. Decide how the code should transition into and out of that state from all other states, and make all necessary updates to the code on the first page.